

Demo Web Shop Capstone Testing Plan

QA automation capstone scope document using Playwright with JavaScript

1. Project Overview

This capstone validates important customer-facing e-commerce flows on Demo Web Shop using Playwright Test with JavaScript. The project uses reusable helpers, service-wise test files, data-driven scenarios, Playwright reporting, Allure reporting, and GitHub Pages hosting.

Field	Details
Website under test	Demo Web Shop
Website URL	https://demowebshop.tricentis.com/
Project type	E-commerce functional UI automation
Automation framework	Playwright Test with JavaScript
Reporting	Playwright HTML report and Allure report
CI/CD	GitHub Actions with GitHub Pages report hosting
Minimum requirement	8 services, 15 test cases per service, total 120 test cases
Cross-browser execution	Chromium, Firefox, and WebKit

2. Testing Objective

- Verify major e-commerce pages and guest-user flows.
- Cover at least 8 services and 120 automated test cases.
- Execute the same 120 scenarios on Chromium, Firefox, and WebKit.
- Generate a combined Allure report for trainer review through GitHub Pages.
- Keep testing stable by focusing on public guest-user validations.

3. Scope Of Testing

In Scope	Out Of Scope
Guest login/register page validation	Real customer credentials or private accounts
Catalog, category, product, cart, wishlist, compare, support, and checkout page validations	Real payment processing or live order placement
Public page UI assertions and navigation checks	Backend database or server-side testing
Cross-browser UI automation	Load, security, and performance testing

4. Service-Wise Test Coverage

Service	Test Cases	Coverage Focus
Authentication	15	Login, register, password recovery, guest account protection, header access links
Product and Search	15	Home page, categories, product details, search results, sorting controls
Cart	15	Empty cart, cart progress, add-to-cart controls, cart links, footer continuity
Wishlist and Compare	15	Wishlist, compare products, recently viewed, email-a-friend controls
User Profile and Account	15	Guest account redirects, register fields, account footer links, account recovery
Address and Shipping	15	Shipping returns, address redirects, delivery/availability text, checkout progress
Checkout and Payment	15	Cart checkout steps, gift card payment fields, product price validation
Customer Support and Information	15	Contact us, sitemap, privacy, conditions, about, news, blog, social links

Total functional scenarios: 120. Since the suite runs on 3 browsers, the final report shows 360 browser executions.

5. Test Case Design Approach

- Each service has one Playwright spec file under tests/services.
- Each service imports scenarios from tests/data/serviceScenarios.js.
- The shared runner opens the route and applies title, heading, text, visibility, enabled state, and count checks.
- Scenario names use IDs such as AUTH-001, PROD-001, CART-001, and INFO-001 for report clarity.

6. Browser Strategy

Browser Project	Expected Executions	Purpose
Chromium	120	Chrome/Edge-like rendering and behavior
Firefox	120	Mozilla browser behavior
WebKit	120	Safari-like rendering behavior

7. Project File Structure

File/Folder	Purpose
.github/workflows/allure-report.yml	GitHub Actions workflow for CI/CD and GitHub Pages deployment.
tests/data/serviceScenarios.js	Central data file containing 8 services and 120 scenario definitions.
tests/services/*.spec.js	Service-wise Playwright spec files that execute the scenario data.
tests/support/demoWebShop.js	Shared navigation helper for Demo Web Shop pages.
tests/support/serviceRunner.js	Reusable assertion runner used by all service tests.
playwright.config.js	Playwright browser, reporter, timeout, and base URL configuration.
package.json	Node scripts and project dependencies.
README.md	GitHub project summary and execution information.

8. Execution And Reporting Plan

- Install dependencies using npm install or npm ci.
- Run the full suite with npx playwright test.
- Generate the Allure report from allure-results.
- Open the local Allure report when needed.
- Push to GitHub so the workflow publishes the latest report to GitHub Pages.

9. Current Verified Result

Metric	Result
Functional scenarios	120
Browser executions	360
Chromium	120 passed
Firefox	120 passed
WebKit	120 passed
Overall status	360 passed, 0 failed, 0 skipped
Hosted report	https://ajaybhatnagar1712.github.io/Wipro_SDET_Playwright_Capstone/